

CV

Sergio Daniel Hernandez Charpak
 Current position(s): Junior researcher / Postdoc
 Academic age: 1 year(s) 7 month(s)

Education

Degree	Organisation	Duration
PhD / Dr.: Electrical Engineering Grégoire Courtine	EPF Lausanne - EPFL, CH SV INX-SV UPCOURTINE	09.2020 - 02.2025 4 year(s) 6 month(s)
Master: Computational Science and Engineering Dimitri Van De Ville & Grégoire Courtine	EPF Lausanne - EPFL, CH Mathematics	09.2017 - 02.2020 2 year(s) 6 month(s)
Bachelor: Physics Jaime Forero	Universidad de Los Andes, CO Science	01.2010 - 03.2017 7 year(s) 3 month(s)
Bachelor: Systems and Computing Engineering Marcela Hernandez	Universidad de Los Andes, CO Engineering	01.2010 - 03.2017 7 year(s) 3 month(s)
Intermediate Japanese Student	Kyoto Institute of Culture and Language, JP Kyoto Institute of Culture and Language	10.2013 - 03.2014 6 month(s)

Employment

Role	Organisation	Duration
Research associate / Scientific collaborator Jocelyne Bloch & Grégoire Courtine	Centre hospitalier universitaire vaudois - CHUV, CH Department of Clinical Neuroscience	10.2026 - 01.2027 4 month(s)

Role	Organisation	Duration
Junior researcher / Postdoc Grégoire Courtine & Jocelyne Bloch	EPF Lausanne - EPFL, CH EPFL SV INX-SV UPCOURTINE	02.2025 - 08.2026 1 year(s) 7 month(s)
Doctoral student / PhD student Grégoire Courtine	EPF Lausanne - EPFL, CH EPFL SV INX-SV UPCOURTINE	09.2020 - 02.2025 4 year(s) 6 month(s)
Scientific Assistant	EPF Lausanne - EPFL, CH EPFL SV INX-SV UPCOURTINE	02.2020 - 09.2020 8 month(s)
Intern	Nagra Kudelski Group, CH Cloud Infra Team	02.2019 - 07.2019 6 month(s)

Major achievements

Achievement 1

Making the human spinal cord accessible to individualized study

My central scientific contribution has been to render the human spinal cord measurable and simulatable at the level of the individual leveraging structural and functional imaging. During my doctorate I developed task-based functional MRI of the lumbosacral and cervical spinal cord, building dedicated acquisition and analysis pipelines that resolve muscle-specific activity along the cord [1, 2, 3]. In parallel I created a semi-automatic framework that reconstructs each participant's spinal anatomy from MRI and CT, including a rootlet-based identification of individual spinal levels that remains robust in the presence of scoliosis, and used it to build personalized anatomical and biophysical models of the cord [4, 5, 6]. My role spanned the full arc from regulatory groundwork and sequence development through software, acquisition, and analysis. Together these methods open a route to studying the sensorimotor organisation of the human spinal cord non-invasively, and to tailoring interventions to a person's very own anatomy.

[1] journal-article. Hernandez-Charpak, S. D., Kinany, N., Ricchi, I., Schlienger, R., Mattera, L., Martuzzi, R., Nazarian, B., Demesmaeker, R., Rowald, A., Kavounoudias, A., Bloch, J., Courtine, G., & Van De Ville, D. (2025). Towards personalized mapping through lumbosacral spinal cord task fMRI. *Imaging Neuroscience*, 3, imag_a_00455. https://doi.org/10.1162/imag_a_00455. [DOI](#).

[2] journal-article. Schlienger, R., Landelle C., Hernandez-Charpak S. D., Pinzon-Corredor D.M., Caron-Guyon J., Sein J., Nazarian B., Bloch J., Felician O., Anton J.L., Courtine G., Kavounoudias A.. 2025. " Mapping Human Proprioceptive Projections of Upper Limb Muscles Through Spinal Cord fMRI." *Human Brain Mapping* 46, no. 15: e70386. <https://doi.org/10.1002/hbm.70386>. [DOI](#).

[3] journal-article. Rowald, A., Komi, S., Demesmaeker, R., Baaklini, E., Hernandez-Charpak, S. D., Paoles, E., Montanaro, H., Cassara, A., Becce, F., Lloyd, B., Newton, T., Ravier, J., Kinany, N., D'Ercole, M., Paley, A., Hankov, N., Varescon, C., McCracken, L., Vat, M., ... Courtine, G. (2022). Activity-dependent spinal cord neuromodulation rapidly restores trunk and leg motor functions after complete paralysis. *Nature Medicine*, 28(2), 260–271. <https://doi.org/10.1038/s41591-021-01663-5>. [DOI](#).

- [4] journal-article. Milekovic, T., Moraud, E.M., Macellari, N. et al. A spinal cord neuroprosthesis for locomotor deficits due to Parkinson's disease. *Nat Med* 29, 2854–2865 (2023). <https://doi.org/10.1038/s41591-023-02584-1>. [DOI](#).
- [5] journal-article. Phillips, A.A., Gandhi, A.P., Hankov, N., Hernandez-Charpak, S.D., et al. An implantable system to restore hemodynamic stability after spinal cord injury. *Nat Med* 31, 2946–2957 (2025). <https://doi.org/10.1038/s41591-025-03614-w>. [DOI](#).
- [6] preprint. Gregoire Courtine, Thibaut Collin, Bianca Yu, Icare Sakr, Sergio D. Hernandez-Charpak et al. An Implantable Digital Bridge Enables Arm And Hand Movements After Tetraplegia, 10 June 2026, PREPRINT (Version 1) available at Research Square [<https://doi.org/10.21203/rs.3.rs-9901682/v1>]. [DOI](#).
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Achievement 2

Neuroimaging and personalized modelling that guide function-restoring neuroprostheses

I have contributed the imaging-and-modelling foundations that let implanted neuroprostheses be tailored to the individual. Working within clinical teams, I built the personalized anatomical models used to plan where epidural stimulation arrays are placed and how they are programmed, together with the brain and spinal imaging used to target them. These models supported proof-of-concept interventions that restored walking [1], regulated blood pressure after spinal cord injury and in multiple system atrophy [2, 3], and re-established upper-limb control [4]. I prepared the models across a series of participants and surgeries, and translated the same imaging-to-model pipeline into an industry collaboration advancing these therapies in close collaboration with Onward Medical. My specific contribution was the quantitative bridge from each person's anatomy and physiology to a concrete, individualized surgery plan, contributing into making such interventions more precise and reproducible.

- [1] journal-article. Milekovic, T., Moraud, E.M., Macellari, N. et al. A spinal cord neuroprosthesis for locomotor deficits due to Parkinson's disease. *Nat Med* 29, 2854–2865 (2023). <https://doi.org/10.1038/s41591-023-02584-1>. [DOI](#).
- [2] journal-article. Phillips, A.A., Gandhi, A.P., Hankov, N., Hernandez-Charpak, S.D., et al. An implantable system to restore hemodynamic stability after spinal cord injury. *Nat Med* 31, 2946–2957 (2025). <https://doi.org/10.1038/s41591-025-03614-w>. [DOI](#).
- [3] journal-article. Squair, J. W., Berney, M., Jimenez, M. C., Hankov, N., Demesmaeker, R., Amir, S., Paley, A., Hernandez-Charpak, S., Dumont, G., Asboth, L., Allenbach, G., Becce, F., Schoettker, P., Wuerzner, G., Bally, J. F., Courtine, G., & Bloch, J. (2022). Implanted System for Orthostatic Hypotension in Multiple-System Atrophy. *New England Journal of Medicine*, 386(14), 1339–1344. <https://doi.org/10.1056/nejmoa2112809>. [DOI](#).
- [4] preprint. Gregoire Courtine, Thibaut Collin, Bianca Yu, Icare Sakr, Sergio D. Hernandez-Charpak et al. An Implantable Digital Bridge Enables Arm And Hand Movements After Tetraplegia, 10 June 2026, PREPRINT (Version 1) available at Research Square [<https://doi.org/10.21203/rs.3.rs-9901682/v1>]. [DOI](#).
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Achievement 3

Interdisciplinary range, mentoring, and translational engagement

My trajectory reflects a deliberate movement across disciplines and a commitment to building capacity in others. I began in physics, contributing to experimental and observational astrophysics before moving into computational science and then neuroengineering; carrying quantitative methods across these

fields has shaped how I frame problems. As Imaging and Computation Manager, I now coordinate imaging and computational work across projects and teams. I have supervised more than twenty master's, semester, and internship students across several institutions, and contributed to teaching, including courses in which students develop assistive technologies together with people with disabilities, work I regard as central to the purpose of the field. I also engage directly with industry to move laboratory methods toward real-world therapies, and I share methods and data openly to support reuse by the wider research community.
